

Effect Size from Claims

Table 1 presents the effect sizes associated with each claim, organized according to the results section of the manuscript and the research questions. The table shows both the effect size and the statistical test applied for its determination. For the Kruskal-Wallis test, we calculate the effect size using the η^2 (eta squared) metric, derived from the H-statistic as follows:

$$\eta^2 = \frac{H - k + 1}{n - k}$$

In this equation, H represents the H-statistic from the Kruskal-Wallis test, k denotes the number of distinct groups, and n is the total number of observations. The η^2 value, ranging from 0 to 1, when multiplied by 100, indicates the percentage of variance in the dependent variable that is attributable to the independent variable.

For models encompassing more than two groups, the partial η^2 is employed to measure the variance proportion attributed to a specific variable, exclusive of the variance accounted for by other variables within the model. The computation of partial η^2 is given by:

$$Partial \eta^2 = \frac{SS_{effect}}{SS_{effect} + SS_{error}}$$

where SS_{effect} is the sum of squares due to the effect of a single variable, and SS_{error} is the sum of squares error within the ANOVA model. The partial η^2 values also span from 0 to 1, with values nearing 1 signifying a greater proportion of variance explained by the variable in question, after considering the variance explained by other variables in the model.

Common interpretation thresholds for both η^2 and partial η^2 , as frequently cited in literature, are as follows: values from 0.01 to less than 0.06 indicate a small effect, values from 0.06 to less than 0.14 suggest a moderate effect, and values of 0.14 or higher reflect a large effect.

Furthermore, we employ Cohen's d for pairwise comparisons, which represents a standardized measure of the difference between two means. Cohen's d is extensively utilized in meta-analyses to quantify the effect size between groups. Cohen's d ranges from 0 to 2 with values around 0.5 suggesting a medium effect and values greater than 0.8 considered as a large effect.

RQ	Claim	Description	Test Used	Statistic	p-value	Effect Size	Effect Size Test
1	Significant differences in perceived usefulness between groups	Post-survey analysis on perceived usefulness	Kruskal-Wallis	14.30	0.006	0.06	η^2
1	Significant differences in behavioral intention between groups	Post-survey analysis on behavioral intention	Kruskal-Wallis	14.20	0.007	0.06	η^2
1	G_R^A perceived the tool as more useful than G_R^{NA}	Pairwise comparison on perceived usefulness	Wilcoxon Rank Sum	2.73	0.006	0.69	Cohen's d
1	G_R^A perceived the tool as more useful than G_{NR}^{NA}	Pairwise comparison on perceived usefulness	Wilcoxon Rank Sum	2.78	0.005	0.60	Cohen's d
1	G_R^A perceived the tool as more useful than CG	Pairwise comparison on perceived usefulness	Wilcoxon Rank Sum	2.75	0.006	0.64	Cohen's d
1	G_{NR}^A reported higher perceived usefulness than G_R^{NA}	Pairwise comparison on perceived usefulness	Wilcoxon Rank Sum	1.96	0.050	0.47	Cohen's d
1	G_R^A exhibited significantly higher behavioral intention scores than G_R^{NA}	Pairwise comparison on behavioral intention	Wilcoxon Rank Sum	2.04	0.041	0.49	Cohen's d
1	G_R^A exhibited significantly higher behavioral intention scores than G_{NR}^{NA}	Pairwise comparison on behavioral intention	Wilcoxon Rank Sum	2.18	0.029	0.49	Cohen's d
1	G_R^A exhibited significantly higher behavioral intention scores than CG	Pairwise comparison on behavioral intention	Wilcoxon Rank Sum	2.04	0.042	0.46	Cohen's d

RQ	Claim	Description	Test Used	Statistic	<i>p</i> -value	Effect Size	Effect Size Test
1	G_{NR}^A exhibited significantly higher behavioral intention scores than G_R^{NA}	Pairwise comparison on behavioral intention	Wilcoxon Rank Sum	2.64	0.008	0.64	Cohen's d
1	G_{NR}^A exhibited significantly higher behavioral intention scores than G_{NR}^{NA}	Pairwise comparison on behavioral intention	Wilcoxon Rank Sum	2.95	0.003	0.65	Cohen's d
1	G_{NR}^A exhibited significantly higher behavioral intention scores than CG	Pairwise comparison on behavioral intention	Wilcoxon Rank Sum	2.44	0.015	0.59	Cohen's d
1	Groups with adaptive feedback had significantly higher scores in perceived usefulness	Comparison between groups with and without adaptive feedback	Kruskal-Wallis	11.46	0.001	0.07	η^2
1	Groups with adaptive feedback had significantly higher scores in attitudes toward use	Comparison between groups with and without adaptive feedback	Kruskal-Wallis	5.28	0.022	0.03	η^2
1	Groups with adaptive feedback had significantly higher scores in behavioral intention	Comparison between groups with and without adaptive feedback	Kruskal-Wallis	12.09	0.001	0.08	η^2
1	Groups with adaptive feedback had significantly higher scores in perceived learning gain	Comparison between groups with and without adaptive feedback	Kruskal-Wallis	6.08	0.014	0.04	η^2
2	Significant effect of test time on quality scores	Repeated measures ANOVA on quality scores	Repeated measures ANOVA	84.40	0.000	0.32	Partial η^2

RQ	Claim	Description	Test Used	Statistic	p-value	Effect Size	Effect Size Test
2	Significant interaction effect on quality scores	Repeated measures ANOVA on quality scores	Repeated measures ANOVA	2.60	0.034	0.06	Partial η^2
2	G_R^A improved more than G_R^{NA}	Pairwise comparison on quality score improvement	Wilcoxon Rank Sum	2.09	0.037	0.59	Cohen's d
2	G_{NR}^A performed better than G_R^{NA}	Pairwise comparison on quality score improvement	Wilcoxon Rank Sum	2.46	0.014	0.53	Cohen's d
2	G_{NR}^A performed better than CG	Pairwise comparison on quality score improvement	Wilcoxon Rank Sum	2.11	0.035	0.49	Cohen's d
2	Significant effect of test time on the transfer task	Repeated measures ANOVA on quality scores	Repeated measures ANOVA	104.00	0.000	0.36	Partial η^2
2	Significant effect of test time on the predicted stars	Repeated measures ANOVA on predicted stars	Repeated measures ANOVA	19.20	0.000	0.10	Partial η^2
3	Groups spent different amount of time revising their recipes	Analysis on the time users revised their recipes	Kruskal-Wallis	12.22	0.016	0.01	η^2
3	users in group G_{NR}^{NA} spent significantly less time revising than users in G_R^A	Pairwise comparison on revision time	Wilcoxon Rank Sum	2.68	0.007	0.29	Cohen's d
3	users in group G_{NR}^{NA} spent significantly less time revising than users in G_{NR}^A	Pairwise comparison on revision time	Wilcoxon Rank Sum	2.88	0.004	0.17	Cohen's d

RQ	Claim	Description	Test Used	Statistic	p-value	Effect Size	Effect Size Test
3	in the first recipe the users in G_R^A spent on average more than twice as much time (346 seconds) as the users in G_R^{NA} (166 seconds, $p < .05$)	Pairwise comparison on revision time	Wilcoxon Rank Sum	2.09	0.036	0.52	Cohen's d
3	Participants from different groups revised their recipes a different number of times	Analysis on number of times users revised their recipes	Kruskal-Wallis	23.68	0.000	0.04	η^2
3	G_{NR}^A revised their recipes more than G_R^A	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	2.29	0.022	0.37	Cohen's d
3	G_{NR}^A revised their recipes more than G_R^{NA}	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	3.80	0.000	0.60	Cohen's d
3	G_{NR}^A revised their recipes more than G_{NR}^{NA}	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	3.68	0.000	0.58	Cohen's d
3	G_{NR}^A revised their recipes more than CG	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	2.69	0.007	0.44	Cohen's d
3	In the first recipe, the users in G_{NR}^A revised their recipe significantly more than G_R^{NA}	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	2.58	0.010	0.69	Cohen's d
3	In the first recipe, the users in G_{NR}^A revised their recipe significantly more than G_{NR}^{NA} .	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	2.65	0.008	0.73	Cohen's d

RQ	Claim	Description	Test Used	Statistic	p -value	Effect Size	Effect Size Test
3	In the second recipe, G_{NR}^A had significantly more revisions than G_{NR}^{NA}	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	2.46	0.014	0.62	Cohen's d
3	In the second recipe, G_{NR}^A had significantly more revisions than G_R^{NA}	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	2.41	0.016	0.66	Cohen's d
3	In the second recipe, G_{NR}^A had significantly more revisions than CG	Pairwise comparison on revision occurrences	Wilcoxon Rank Sum	1.97	0.048	0.53	Cohen's d

Table 1: Details of Claims, Descriptions, Statistical Tests, Results, and Effect Sizes in the Study